

Chapter 13: The "Network" (Intro to APIs)

I. API Fundamentals & CRUD Operations

- **API (Application Programming Interface):** A set of _____ (a "menu") that lets your _____ (browser) communicate with the _____ (kitchen) for data.
- **The Waiter Analogy:** The API is the _____ that takes your **Request** (order) and brings back a **Response** (data).
- **Endpoint:** The specific _____ address you use to ask for a piece of data (defined by the API's _____).

Mastering the Request (CRUD)

The full power of an API is managed through four core **HTTP Methods** that map to the **CRUD** operations.

HTTP Method	CRUD Operation	Purpose (The Action)	Analogy (The Waiter)
GET	_____	_____ data from the server (most common front-end request).	"I'd like the steak, please."
POST	_____	Send data to the server to create a _____ resource (e.g., new user).	"I want to submit my order."
PUT/PATCH	_____	_____ an existing resource (e.g., change username).	"I need to change my side dish."
DELETE	_____	_____ a	"Please take this

		resource from the server (e.g., delete a comment).	plate away."
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Deep Dive Discussion

1. **Front-End Focus:** If you are building a front-end website that only displays a list of news articles, which of the four CRUD methods will you use 99% of the time, and why?
2. **API Standardization:** Why is it crucial that an API follows a "set of rules" (documentation) instead of letting the developer design their own random requests?

II. The Asynchronous Challenge & Data

Solving the Freeze Problem

- **Synchronous Code:** Code that runs _____; the browser would _____ until the request finishes.
- **Asynchronous Code:** Code that does **not** freeze the site. It sends the request and _____ (the main thread can go do other things).
- **fetch():** The modern JavaScript command for making an API request, which returns a _____ object.

The async/await Pattern

- **async:** Marks a function as containing _____ code.
- **await:** Tells JavaScript to "_____ here, but do not block the thread." Execution pauses only for that specific operation.

Data Format

- **JSON (JavaScript Object Notation):** The standard data format sent back by the server. It is structurally identical to JavaScript _____ and _____.
- **Translation:** The crucial command that converts the raw network stream into a usable JS Object is _____.

Application Activity

1. **Async Logic:** Arrange the following actions in the order they happen in a function using async/await to get data from an API:
 - **A.** Translate the raw stream into a JSON object (await response.json()).
 - **B.** Send the request over the network (await fetch(...)).
 - **C.** Display the final data on the page.
 - **Order:** 1. _____ 2. _____ 3. _____

III. Robustness & Error Handling

Status Codes (The Server's Reply)

The server always sends a numerical _____ back with the response.

Code Range	Meaning	Client Action
2xx (e.g., 200 OK)	_____	Proceed to parse the JSON.
4xx (e.g., 404 Not Found)	_____ Error	Stop and handle the error (don't try to read the data).
5xx (e.g., 500 Internal Server Error)	_____ Error	Stop and handle the error.

The Safety Net

- **try / catch:** The block used for professional error handling. The risky await calls are wrapped in the _____ block.
- **Failure Scenario:** If **any** line inside the try block fails (e.g., network failure, a 4xx error), execution immediately jumps to the _____ block.

Deep Dive Discussion

1. **Error Handling Necessity:** Why is a try/catch block more professional than just letting the code fail? (Hint: Think about what the user sees versus what the developer sees).
2. **Status Code Logic:** The server sends back a **404** status code. The request was technically *completed* (the server replied), but it failed. Why should your JavaScript throw an error and execute the catch block instead of trying to read the data?

Application Activity

1. **Debugging Check:** You are debugging an API call using console.log(). If the console shows the message: Promise Sent: Promise {<pending>}, what does this mean, and what word is missing from your code that would fix it?
 - **Meaning:**

 - **Missing Keyword:** _____